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Fastener

Abstract

The present application provides a fastener for fastening a first component in a second component having a circular hole. The fastener includes a main body and a retaining member. The main body includes a head for connecting the first component and a shaft portion extending from the head. The retaining member includes a receiving barrel and a plurality of retaining arms. The shaft portion passes through and is connected to the receiving barrel to limit the axial and radial movements of the retaining member along the shaft portion relative to the main body. The plurality of retaining arms extend upwardly from the receiving barrel and are distributed about the central axis of the receiving barrel. The plurality of retaining arms are deflectable towards the shaft portion. The plurality of retaining arms are located between the shaft portion and a hole and abut against the hole to retain the second component between the head and the plurality of retaining arms when the retaining member is inserted into the hole. The main body is made of a first material, and the retaining member is made of a second material that is more wear-resistant than the first material.

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Background/Summary

RELATED APPLICATION

(1) The present application claims the benefit of Chinese Patent Application Nos. 202310395435X, filed Apr. 13, 2023, titled “Fastener,” the contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present application relates to a fastener. Automotive components require fastening techniques that are simple to manufacture and assemble. Further, fastening techniques should above all be reliable and efficient. In order to secure a secondary panel to a primary panel, a fastener may be used, such as a pin and grommet fastener.

(3) Various types of fasteners are used for fastening components. For example, in case of an instrument panel of a vehicle, fasteners may be used to fix together adjacent panels or to secure one or more objects on the panels. One such type of fasteners is usable with holes of different types, sizes, and shapes provided in the components to be secured together. In other words, for such a fastener to fasten the components, at least one of the components is provided with a hole. One of the components is mounted on the fastener and other component having the hole receives the fastener component assembly.

BACKGROUND

(4) When it is necessary to connect two components, such a connection is achieved by using a fastener. For example, during vehicle assembly, vehicle body trim panels and a vehicle frame are connected by using fasteners.

SUMMARY OF THE DISCLOSURE

(5) The present disclosure generally relates to a fastener, substantially as illustrated by and described in connection with at least one of the figures, as set forth more completely in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other objects, features, and advantages of the devices, systems, and methods described herein will be apparent from the following description of particular examples thereof, as illustrated in the accompanying figures; where like or similar reference numbers refer to like or similar structures. The figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the devices, systems, and methods described herein.

(2) FIG. 1 is a perspective view of a fastener according to an embodiment of the present application.

(3) FIG. 2 is an exploded view of the fastener in FIG. 1.

(4) FIG. 3 is a schematic perspective view illustrating the fastener **10** in FIG. 1 connecting a first component and a second component.

(5) FIG. 4 is a perspective view of a retaining member of the fastener in FIG. 1 in a certain direction.

(6) FIG. 5 is a perspective view of the retaining member in FIG. 4 in another direction.

(7) FIG. 6 is a perspective view of a main body of the fastener in FIG. 1.

(8) FIG. 7 is a schematic perspective view illustrating the fastener in FIG. 1 connected to the second component.

(9) FIG. 8 is a schematic sectional view illustrating the fastener in FIG. 1 connected to the second component, with the section passing through a retaining arm of the retaining member.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) Various specific embodiments of the present application will be described below with reference to the accompanying drawings which form a part of this description. It should be understood that, where possible, the same or similar reference numerals used in the present application refer to the same components.

(11) Various specific embodiments of the present application will be described below with reference to the accompanying drawings which form a part of this description. It should be understood that although the terms such as “upper”, “lower”, “inner”, “outer” and so on indicating directions are used in the present application to describe various illustrative structural parts and elements in the present application, the terms used herein are merely used for case of description and are determined based on the illustrative orientation shown in the accompanying drawings. Since the embodiments disclosed in the present application may be arranged in different directions, these terms indicating directions are merely illustrative and should not be considered as limitations.

(12) The present application provides a fastener formed by connecting two portions, and the two portions of the fastener can be connected securely. The fastener is capable of securely connecting objects and still maintaining a strong fastening to the connected objects after multiple insertions into and pulls out of the connected objects.

(13) The fastener according to the present application is used for fastening a first component to a second component having a circular hole. The fastener comprises a main body and a retaining member. The main body comprises a head configured to connect the first component and a shaft portion extending from the head. The retaining member comprises a receiving barrel and a plurality of retaining arms. The shaft portion passes through and is connected to the receiving barrel to limit the axial and radial movements of the retaining member along the shaft portion relative to the main body, and the plurality of retaining arms extend upwardly from the receiving barrel, are distributed about the central axis of the receiving barrel and are configured to be deflectable towards the shaft portion, wherein the plurality of retaining arms and the shaft portion are configured such that the plurality of retaining arms are located between the shaft portion and the hole and abut against the hole to retain the second component between the head and the plurality of retaining arms when the retaining member is inserted into the hole. The main body is made of a first material, and the retaining member is made of a second material that is more wear-resistant than the first material.

(14) In the fastener as mentioned above, the first material is plastic, and the second material is metal.

(15) In the fastener as mentioned above, the retaining member is removably mounted on the main body.

(16) In the fastener as mentioned above, the shaft portion comprises a shaft body, and an upper stop structure and a lower stop structure disposed on the shaft body. The upper stop structure is spaced apart from the head by a certain distance, and the lower stop structure is located below the upper stop structure, facing the upper stop structure, and is spaced apart from the upper stop structure by a certain distance. The receiving barrel of the retaining member cooperates with the upper stop structure and the lower stop structure to limit the axial movement of the retaining member along the shaft portion relative to the main body, and the receiving barrel cooperates with the portion of the shaft portion between the upper stop structure and the lower stop structure to limit the radial movement of the retaining member along the shaft portion relative to the main body.

(17) In the fastener as mentioned above, the receiving barrel comprises a barrel body. The outer diameter of the portion of the shaft body (112) received in the barrel body matches the inner diameter of the barrel body.

(18) In the fastener as mentioned above, the receiving barrel further comprises a plurality of upper connecting legs extending from the upper end of the barrel body, the plurality of upper connecting legs are arranged about the central axis of the barrel body, and the plurality of upper connecting legs and the plurality of retaining arms are arranged alternately about the central axis of the barrel body. The shaft portion further comprises a plurality of protrusions arranged about the central axis of the shaft portion and protruding outwardly from the shaft body, and the upper stop structure comprises a plurality of upper stop bosses respectively formed by the plurality of protrusions. The plurality of upper connecting legs respectively abut against the plurality of upper stop bosses.

(19) In the fastener as mentioned above, the receiving barrel further comprises a plurality of lower

connecting legs extending from the lower end of the barrel body, and the plurality of lower connecting legs are arranged about the central axis of the barrel body and extend inwardly obliquely towards the central axis of the barrel body. The shaft portion further comprises a plurality of lower recesses arranged about the central axis of the shaft portion and recessed inwardly from the shaft body, and the lower stop structure comprises a plurality of lower stop bosses respectively formed by the plurality of lower recesses. The plurality of lower connecting legs respectively abut against the plurality of lower stop bosses, and the plurality of lower connecting legs are respectively at least partially received by the plurality of lower recesses.

(20) In the fastener as mentioned above, the receiving barrel further comprises a plurality of guiding tabs extending from the lower end of the barrel body, the plurality of guiding tabs are arranged about the central axis of the barrel body and extend inwardly obliquely towards the central axis of the barrel body, and the plurality of guiding tabs and the plurality of lower connecting legs are alternately arranged about the central axis of the barrel body.

(21) In the fastener as mentioned above, the barrel body has a hexagonal cross-section, and the receiving barrel comprises three lower connecting legs, three guiding tabs, three upper connecting legs and three retaining arms, the three lower connecting legs and the three guiding tabs respectively extend downwardly from six sides of the barrel body, and the three upper connecting legs and the three retaining arms respectively extend upwardly from the six sides of the barrel body.

(22) In the fastener as mentioned above, the receiving barrel has a discontinuity enabling the receiving barrel to expand in the radial direction of the receiving barrel.

(23) In the fastener as mentioned above, the shaft portion comprises a plurality of upper ribs located above the upper stop structure, and the plurality of upper ribs are arranged about the central axis of the shaft portion and cooperate with the hole to center the fastener in the hole. The shaft portion further comprises a plurality of receiving recesses formed between adjacent ones of the plurality of upper ribs, and the plurality of receiving recesses are configured to respectively receive the plurality of retaining arms when the plurality of retaining arms are deflected towards the shaft portion.

(24) In the fastener as mentioned above, the plurality of retaining arms have a substantially V shape such that each of the plurality of retaining arms has a lower portion extending obliquely away from the central axis of the barrel body from the barrel body and an upper portion extending obliquely inwardly towards the central axis of the barrel body from the lower portion, and the upper portion has a contact portion for abutting against the hole.

(25) In the fastener as mentioned above, the end of the upper portion is bent outwardly to form a limiting stop retaining the upper portion in the hole.

(26) In the fastener as mentioned above, the plurality of retaining arms have arcuate protrusions on their outer surfaces.

(27) A fastener of the present application is used for connecting a first component **20** and a second component **30** (shown in FIG. 3). In some embodiments, the first component **20** is a vehicle body trim panel, and the second component **30** is a vehicle frame. It should be understood that in other embodiments, the first component **20** and the second component **30** are other types of components that need to be connected to each other.

(28) FIG. 1 is a perspective view of a fastener **10** according to an embodiment of the present application. FIG. 2 is an exploded view of the fastener **10** in FIG. 1. As shown in FIGS. 1 and 2, the fastener **10** comprises a main body **100** and a retaining member **200**. The main body **100** comprises a head **101** and a shaft portion **111** extending from the head **101**, and the retaining member **200** is removably assembled on the shaft portion **111** of the main body **100**. The main body **100** is made of a first material, and the retaining member **200** is made of a second material that is more wear-resistant than the first material. In some embodiments, the first material is plastic, and the second material is metal. It should be understood that in other embodiments, the first material and the

second material are other suitable materials. The fastener **10** further comprises a scaling ring **300** removably connected to the main body **100**.

(29) FIG. **3** is a schematic perspective view illustrating the fastener **10** in FIG. **1** connecting a first component **20** and a second component **30**. As shown in FIG. **3**, the head **101** of the main body **100** of the fastener **10** is connected to the first component **20**, and the retaining member **200** assembled on the shaft portion **111** of the main body **100** passes through a circular hole **31** (shown in FIG. **8**) of the second component **30** to connect the second component **30** between the retaining member **200** and the head **101** of the main body **100**.

(30) FIG. **4** is a perspective view of the retaining member **200** of the fastener **10** in FIG. **1** in a certain direction. The retaining member **200** comprises a receiving barrel **211** and three retaining arms **202**. The receiving barrel **211** comprises a barrel body **212** having a hexagonal cross-section. The receiving barrel **211** comprises three lower connecting legs **216**, three guiding tabs **217** and three upper connecting legs **215**. The three lower connecting legs **216** and the three guiding tabs **217** respectively extending downwardly from the six sides of the lower end **214** of the barrel body **212** are arranged alternately about the central axis of the barrel body **212**, and extend obliquely inwardly towards the central axis of the barrel body **212**. The three lower connecting legs **216** and the three guiding tabs **217** extend obliquely inwardly towards the central axis of the barrel body **212**, such that when the fastener **10** is being inserted into the hole **31** of the second component **30**, the retaining member **200** is guided to slide downwardly against the hole **31** to insert into the hole **31**, that is, the retaining member **200** easily slides downwardly against the hole **31** to insert into the hole **31**. The three upper connecting legs **215** extend vertically upwardly from three sides of the upper end **213** of the barrel body **212** and are arranged about the central axis of the barrel body **212**. The three upper connecting legs **215** and the three guiding tabs **217** respectively extend upwardly and downwardly from opposite sides of the upper end **213** and the lower end **214** of the barrel body **212**. In some embodiments, the ends of the upper connecting legs **215** are inclined away from the central axis of the barrel body **212** (as shown in FIG. **4**). In other embodiments, the ends of the upper connecting legs **215** extend vertically upward.

(31) The configuration of the three retaining arms **202** will be described with continuing to refer to FIG. **4**. The three retaining arms **202** extending upwardly from the receiving barrel **211** are distributed about the central axis of the receiving barrel **211** (coincide with the central axis of the barrel body **212**), and are configured as deflectable towards the central axis of the receiving barrel **211**. The three retaining arms **202** and the three upper connecting legs **215** are alternately arranged about the central axis of the barrel body **212**. The retaining arms **202** each has a substantially V shape such that each of the retaining arms **202** has a lower portion **203** and an upper portion **204**. The lower portions **203** extend obliquely away from the central axis of the barrel body **212** from the barrel body **212**, and the upper portions **204** extend obliquely inwardly towards the central axis of the barrel body **212** from the lower portions **203**. When the fastener **10** is connected to the second component **30**, the retaining arms **202** are located between the shaft portion **111** of the main body **100** and the hole **31** of the second component **30**, and the upper portions **204** abut against the hole **31** at a contact portion **220** (see FIG. **8**), so as to retain the second component **30** between the head **101** of the main body **100** and the retaining arms **202** of the retaining member **200**. The upper portions **204** are bent outwardly to form limiting stops **205** which retain the upper portions **204** in the hole **31** when the fastener **10** is connected to the second component **30**. As shown in FIG. **4**, each retaining arm **202** has an arcuate protrusion **206** on the outer surface thereof, which cooperates with a surface of the hole **31** to enable the retaining arm **202** to easily slide against the hole **31** so as to facilitate inserting or pulling out the fastener **10** into/from the hole **31** in the process of inserting or pull out the fastener **10** into/from the hole **31** of the second component **30**.

(32) Although it is showed in the figures that the barrel body **212** of the receiving barrel **211** has a hexagonal cross-section, and correspondingly having three upper connecting legs **215**, three lower connecting legs **216**, three guiding tabs **217** and three retaining arms **202**, in some embodiments,

the barrel body **212** is designed to have cross-sections of other shapes, and having different numbers of upper connecting legs **215**, lower connecting legs **216**, guiding tabs **217** and retaining arms **202** that extend from the barrel body **212**.

(33) FIG. 5 shows a perspective view of the retaining member **200** in FIG. 4 in another direction. As shown in FIG. 5, the receiving barrel **211** has a discontinuity **209** which enables the receiving barrel **211** to expand in the radial direction of the receiving barrel **211**, allowing the shaft portion **111** of the main body **100** to be easily inserted into the receiving barrel **211**. The discontinuity **209** also allows that, in some embodiments of manufacturing the retaining member **200**, the sheet-shaped retaining member **200** is first manufactured, and the sheet-shaped retaining member **200** is then rolled to have a barrel shape as shown in the figures of the present application. After the sheet-shaped retaining member **200** is rolled to have the barrel shape, the discontinuity **209** is formed between the two ends of the sheet-shaped retaining member **200**. Such an implementation makes it easy to manufacture the retaining member **200**.

(34) FIG. 6 is a perspective view of the main body **100** of the fastener **10** in FIG. 1. As shown in FIG. 6, the head **101** of the main body **100** comprises a disk **102** and a disk **103** that are connected to each other. The disk **103** is located under the disk **102**, facing the disk **102**, and is spaced apart from the disk **102** by a gap **104**. The first component **20** has an opening (not shown in the figure) that cooperates with the connection of the disk **102** and the disk **103** such that the first component **20** can be connected between the disk **102** and the disk **103**. When the fastener **10** is connected to the first component **20**, the first component **20** is connected to the head **101** with being retained in the gap **104**. When the fastener **10** is further connected to the second component **30**, the first component **20** and the second component **30** are connected by means of the fastener **10**. The sealing ring **300** wraps the periphery of the disk **103** and extends beyond the lower surface of the disk **103**. When the fastener **10** is connected to the second component **30**, the sealing ring **300** abuts against the upper surface of the second component **30** to form a seal (refer to FIGS. 1 and 8).

(35) Continuing to refer to FIG. 6, the shaft portion **111** extends from the disk **103**. The shaft portion **111** comprises a shaft body **112**. The outer diameter of a portion of the shaft body **112** received in the barrel body **212** matches the inner diameter of the barrel body **212**. The shaft body **112** is provided with an upper stop structure **113** and a lower stop structure **115**. The upper stop structure **113** is spaced apart from the disk **103** by a certain distance. The lower stop structure **115** is located below the upper stop structure **113**, facing the upper stop structure **113**, and is spaced apart from the upper stop structure **113** by a certain distance. When the retaining member **200** is assembled on the main body **100**, the shaft portion **111** of the main body **100** passes through the receiving barrel **211** and is connected to the receiving barrel **211** to limit the axial and radial movements of the retaining member **200** along the shaft portion **111** relative to the main body **100**. Specifically, when the retaining member **200** is assembled on the main body **100**, the receiving barrel **211** cooperates with the upper stop structure **113** and the lower stop structure **115** to limit the axial movement of the retaining member **200** along the shaft portion **111** relative to the main body **100**. Moreover, the receiving barrel **211** cooperates with a portion of the shaft portion **111** located between the upper stop structure **113** and the lower stop structure **115** to limit the radial movement of the retaining member **200** along the shaft portion **111** relative to the main body **100**. The receiving barrel **211** can cooperate with the portion of the shaft portion **111** located between the upper stop structure **113** and the lower stop structure **115** to limit the radial movement of the retaining member **200** along the shaft portion **111** relative to the main body **100**.

(36) Further, as shown in FIG. 6, the upper stop structure **113** comprises three upper stop bosses **114** respectively formed by three protrusions arranged about the central axis of the shaft portion **111** and protruding outwardly from the shaft body **112**. The shaft portion **111** comprises three lower recesses **117** arranged about the central axis of the shaft portion **111** and recessed inwardly from the shaft body **112**. The lower stop structure **115** comprises three lower stop bosses **116** respectively formed by three lower recesses **117**. When the retaining member **200** is assembled on the main

body **100**, the three upper connecting legs **215** of the retaining member **200** respectively abut against the three upper stop bosses **114**, and the three lower connecting legs **216** of the retaining member **200** respectively abut against the three lower stop bosses **116**, and thus the axial movement of the retaining member **200** along the shaft portion **111** relative to the main body **100** is limited. Since the three lower connecting legs **216** of the retaining member **200** extend obliquely inwardly towards the central axis of the barrel body **212**, when the retaining member **200** are moving downwardly along the shaft portion **111** relative to the main body **100**, the three lower connecting legs **216** are further bent upwardly so as to increase the contact areas between the lower connecting legs **216** and the corresponding lower stop bosses **116**, and thus the downward movement of the retaining member **200** along the shaft portion **111** relative to the main body **100** can be better limited. In an embodiment in which the ends of the three upper connecting legs **215** are bent outwardly, when the retaining member **200** are moving upwardly along the shaft portion **111** relative to the main body **100**, surface contacts can be formed between the three upper connecting legs **215** and the corresponding upper stop bosses **114**, and thus the upward movement of the retaining member **200** along the shaft portion **111** relative to the main body **100** can be better limited.

(37) In addition, as shown in FIG. 6, when the retaining member **200** is assembled on the main body **100**, the three lower connecting legs **216** of the retaining member **200** are respectively at least partially received by the three lower recesses **117**. Further, as shown in FIG. 6, the lower ribs **118** are formed between adjacent lower recesses **117**. The configuration that the three lower connecting legs **216** of the retaining member **200** are respectively at least partially received by the three lower recesses **117** enables the lower ribs **118** to stop the corresponding lower connecting legs **216** received in the three lower recesses **117** in the radial direction of the shaft portion **111**, which is thus conducive to limiting the radial movement of the retaining member **200** along the shaft portion **111** relative to the main body **100**. Moreover, the configuration that the three lower connecting legs **216** of the retaining member **200** are respectively at least partially received by the three lower recesses **117** allows the three lower connecting legs **216** to remain extending obliquely inwardly towards the central axis of the barrel body **212** when the retaining member **200** is assembled to the main body **100**. Accordingly, when the fastener **10** is being inserted into the hole **31** of the second component **30**, the three lower connecting legs **216** guide the retaining member **200** to slide downwardly against the hole **31** to insert into the hole **31**.

(38) Further referring to FIG. 6, the lower portions of the lower rib **118** has a cut portion **119** which comprises an inclined cutting surface **122** extending obliquely inwardly towards the central axis of the shaft portion **111**. The inclined cutting surfaces **122** are configured to match the guiding tabs **217** of the retaining member **200**. In this connection, when the retaining member **200** is assembled to the main body **100**, the guiding tabs **217** remain extending obliquely inwardly towards the central axis of the barrel body **212** with abutting against the corresponding inclined cutting surfaces **122**. Accordingly, when the fastener **10** is being inserted into the hole **31** of the second component **30**, the guiding tabs **217** guides the retaining member **200** to slide downwardly against the hole **31** to insert into the hole **31**. In addition, the configuration that the guiding tabs **217** abut against the corresponding inclined cutting surfaces **122** is also conducive to limiting the radial movement of the retaining member **200** along the shaft portion **111** relative to the main body **100**.

(39) In the embodiment shown in FIG. 6, the three lower recesses **117** extend upwardly from the lower stop bosses **116** by part of the length of the shaft portion **111**, such that the upper ends of the three lower recesses **117** define the boss **125** between the upper stop structure **113** and the lower stop structure **115**. The shaft portion **111** further comprises three middle recesses **124** respectively aligned with the three lower recesses **117** and extending from the boss **125** to the position where the upper stop structure **113** is located. The lower rib **118** are formed between the adjacent lower recesses **117**, and the middle ribs **123** are formed between adjacent middle recesses **124**. In the embodiment shown in FIG. 6, the middle rib are **123** are aligned to and connected to the lower ribs

118. In other embodiments, the middle ribs **123** are not aligned to and not connected to the lower ribs **118**. In the embodiment shown in FIG. 6, the diameter of the boss **125**, the diameter of the circumscribed circle of the three middle ribs **123** and the diameter of the circumscribed circle of the uncut portions of the three lower ribs **118** are all equal to the diameter of the shaft body **112**. When the retaining member **200** is assembled on the main body **100**, the barrel body **212** wraps the boss **125**, portions of the middle ribs **123** and the uncut portions of the lower ribs **118**, and the upper connecting legs **215** abut against portions of the corresponding middle ribs **123**. Such configuration is conducive to limiting the axial and radial movements of the retaining member **200** along the shaft portion **111** relative to the main body **100**. In the embodiment shown in the figures, the three lower ribs **118** are connected to the boss **125** and the corresponding three lower stop bosses **116**, and the three middle ribs **123** are connected to the boss **125** and the shaft body **112** at the position where the corresponding three upper stop bosses **114** are located, so that the three middle ribs **123** and the three lower ribs **118** are conducive to strengthening the structure of the main body **100**. In some embodiments, the three middle ribs **123** are not connected to the boss **125** and the shaft body **112** at the position where the corresponding three upper stop bosses **114** are located, and the three lower ribs **118** are not connected to the boss **125** and the corresponding three lower stop bosses **116**.

(40) The shaft portion **111** further comprises three upper ribs **120** located above the upper stop structure **113** and three receiving recesses **121** formed between the adjacent ones of the upper ribs **120**. The three upper ribs **120** are arranged about the central axis of the shaft portion **111** and are configured to cooperate with the hole **31** of the second component, such that the three upper ribs **120** center the fastener **10** in the hole **31** when the fastener **10** is connected to the second component **30**. In the embodiment shown in the figures, the three upper ribs **120** are configured to have a circumscribed circle having a diameter slightly less than the diameter of the hole **31**. In this way, the three upper ribs **120** center the fastener **10** in the hole **31**. As shown in the figures, the three upper ribs **120** are aligned to the three middle ribs **123** and the three lower ribs **118**, respectively, and the three receiving recesses **121** are aligned to the three middle recesses **124** and the three lower recesses **117**, respectively. In other embodiments, the three upper ribs **120** are not aligned to the corresponding middle ribs **123** and the corresponding lower ribs **118**, and the three receiving recesses **121** are not aligned to the corresponding middle recesses **124** and the lower recesses **117**.

(41) When the retaining member **200** is assembled on the main body **100**, the three elastic arms **202** of the retaining member **200** are respectively located outside of the corresponding middle recesses **124** and the receiving recesses **121**. Since the three elastic arms **202** are located outside of the corresponding receiving recesses **121**, the three receiving recesses **121** provide spaces for the corresponding retaining arms **202** to be deflected towards the shaft portion **111**, such that the three retaining arms **202** can be deflected towards the shaft portion **111** and received by the corresponding receiving recesses **121** when the fastener **10** is being inserted into the hole **31** of the second component **30**. Meanwhile, the middle recesses **124** are also conducive to preventing the shaft portion **111** from interfering with the deflection of the three elastic arms **202** towards the shaft portion **111**. In the embodiment shown in FIG. 6, the three upper ribs **120** are connected to the upper stop bosses **114** and the lower disk **103** to strengthen the structure of the main body **100**. In some embodiments, the three upper ribs **120** are disposed between the upper stop bosses **114** and the lower disk **103**, but are not connected to the corresponding upper stop bosses **114** and the lower disk **103**. Furthermore, although it is shown in the figures that three lower stop bosses **116**, three lower recesses **117**, three lower ribs **118**, three middle recesses **124**, three middle ribs **123**, three upper stop bosses **114**, three upper ribs **120** and three receiving recesses **121** are provided, in other embodiments, the shaft portion **111** is configured to have different numbers of these structures so as to match the barrel bodies **212** having different cross-sectional shapes.

(42) FIG. 7 is a schematic perspective view illustrating the fastener **10** in FIG. 1 connected to the

second component **30**. For the purpose of clearly showing the cooperation between the retaining arms **202** and the hole **31** of the second component **30**, the sealing ring **300** is not shown in FIG. 7. FIG. 8 is a schematic sectioned view illustrating the fastener **10** in FIG. 1 connected to the second component **30**, with the section passing through one retaining arm **202** of the retaining member **200**. With reference to FIGS. 7 and 8, the connecting and disconnecting of the fastener **10** and the second component **30** are described.

(43) The diameter of the hole **31** of the second component **30** is greater than the outer diameter of the barrel body **212** of the retaining member **200**, such that the barrel body **212** of the fastener **10** and the structures below the barrel body **212** can pass through the hole **31**. After the barrel body **212** is inserted through the hole **31**, the lower portions **203** of the retaining arms **202** come into contact with the hole **31** and slide downwardly against the hole **31** as the fastener **10** is being further inserted into the hole **31**. In this process, the retaining arms **202** are subjected to a force towards the shaft portion **111**, causing the retaining arms **202** to be deflected towards the shaft portion **111** from their initial positions, so that the fastener **10** can continue to be inserted downwardly. When the connections of the lower portions **203** and the upper portions **204** of the retaining arms **202** abut against the hole **31**, the retaining arms **202** deflect the maximum towards the shaft portion **111**. As the fastener **10** continues to be inserted downwardly, the upper portions **204** of the retaining arms **202** come into contact with the hole **31** and slide downwardly against the hole **31**. Thus, the retaining arms **202** are gradually deflected away from the shaft portion **111** towards their initial positions. When the fastener **10** is inserted downwardly to the extent that the sealing ring **300** of the fastener **10** abuts against the second component **30**, the fastener **10** is inserted into the hole **31** in place. At this moment, the retaining arms **202** at least partially return to their initial positions, and the retaining arms **202** are located between the shaft portion **111** and the hole **31** with the contact portions **220** on the upper portions **204** of the retaining arms **202** abut against the hole **31**. Also at this moment, the limiting stops **205** at the ends of the upper portions **204** of the retaining arms **202** extend beyond the upper surface of the hole **31**, such that the limiting stops **205** can retain the upper portions **204** in the hole **31**. Thus, the second component **30** is connected to the fastener **10**.

(44) The fastener **10** can be pulled upwardly when it is necessary to disconnect the fastener **10** from the second component **30**. When the fastener **10** is being pulled upwardly, the upper portions **204** of the retaining arms **202** slide upwardly against the hole **31**, and the retaining arms **202** are subjected to the force towards the shaft portion **111** so as to be deflected towards the shaft portion **111**. When the fastener **10** is pulled upwardly to the extent that the connections of the lower portions **203** and the upper portions **204** of the retaining arm **202** abut against the hole **31**, the retaining arms **202** deflect the maximum towards the shaft portion **111**. As the fastener **10** continues to be pulled upwardly, the lower portions **203** of the retaining arms **202** reach contact with the hole **31** and slide upwardly against the hole **31**. Then, as the lower portions **203** of the retaining arms **202** slide upwardly against the hole **31**, the retaining arms **202** are gradually deflected away from the shaft portion **111** towards their initial positions. When the lower portions **203** slide upwardly to the extent that the retaining arms **202** leave the hole **31**, the retaining arms **202** return to their initial positions. The fastener **10** is removed from the hole **31** with being further pulled upwardly. Thus, the second component **30** is disconnected from the fastener **10**.

(45) Although the retaining arms of the retaining member **200** slide against the hole **31** in the process of inserting and pulling out the fastener **10** into/from the hole **31** of the second component **30**, the retaining arms **202** are not easily to be cut since the retaining member **200** is made of a wear-resistant material. Therefore, the fastener **10** according to the present application can still maintain a strong fastening to the connected second component **30** so as to stably connect the first component **20** and the second component **30** after multiple insertions into and pulls out of the hole **31** of the second component **30**.

(46) The present application offers at least the following technical effects. First, since the fastener

is composed of two portions, the retaining member **200** can be made of a more wear-resistant material. Therefore, the retaining arms **202** are less prone to be cut when the fastener **10** is being inserted into the hole **31** of the second component **30**. Therefore, the fastener **10** according to the present application can still maintain a strong fastening to the connected second component **30** after multiple insertions into and pulls out of the hole **31** of the second component **30**. Second, the configuration of the main body **100** and the retaining member **200** of the fastener **10** enables the fastener **10** to securely connect the second component **30**. Finally, the configuration of the main body **100** and the retaining member **200** of the fastener **10** enables the main body **100** and the retaining member **200** to be securely connected to each other.

(47) Although the present application is described with reference to the examples of embodiments outlined above, various alternatives, modifications, variations, improvements and/or substantial equivalents, which are known or anticipated at present or to be anticipated before long, may be obvious to those of at least ordinary skill in the art. Furthermore, the technical effects and/or technical problems described in this description are exemplary rather than limiting. Therefore, the disclosure in this description may be used to solve other technical problems and have other technical effects and/or may solve other technical problems. Accordingly, the examples of the embodiments of the present disclosure as set forth above are intended to be illustrative rather than limiting. Various changes may be made without departing from the spirit or scope of the present disclosure. Therefore, the present disclosure is intended to embrace all known or earlier disclosed alternatives, modifications, variations, improvements and/or substantial equivalents.

Claims

1. A fastener for fastening a first component in a second component having a circular hole, the fastener comprising: a main body comprising a head configured to connect the first component and a shaft portion extending from the head; and a retaining member comprising a receiving barrel and a plurality of retaining arms extending upwardly from the receiving barrel and distributed about a central axis of the receiving barrel, the shaft portion passing through and connected to the receiving barrel to limit axial and radial movement of the retaining member along the shaft portion relative to the main body, the plurality of retaining arms being configured to be deflectable toward the shaft portion, wherein the plurality of retaining arms and the shaft portion are configured such that the plurality of retaining arms are located between the shaft portion and the hole and abut against the hole to hold the second component between the head and the plurality of retaining arms when the retaining member is inserted into the hole; wherein the main body is made of a first material and the retaining member is made of a second material that is more wear-resistant than the first material.
2. The fastener according to claim 1, wherein the first material is plastic and the second material is metal.
3. The fastener according to claim 2, wherein the retaining member is removably assembled on the main body.
4. The fastener according to claim 1, wherein the shaft portion comprises a shaft body and an upper stop structure and a lower stop structure disposed on the shaft body, wherein the upper stop structure is spaced apart from the head, and the lower stop structure is located below the upper stop structure with facing the upper stop structure and being spaced apart from the upper stop structure; and wherein the receiving barrel of the retaining member cooperates with the upper stop structure and the lower stop structure to limit the axial movement of the retaining member along the shaft portion relative to the main body, and wherein the receiving barrel cooperates with the portion of the shaft portion between the upper stop structure and the lower stop structure to limit the radial movement of the retaining member along the shaft portion relative to the main body.
5. The fastener according to claim 4, wherein the receiving barrel comprises a barrel body, wherein

an outer diameter of the portion of the shaft body received in the barrel body matches an inner diameter of the barrel body.

6. The fastener according to claim 5, wherein the receiving barrel further comprises a plurality of upper connecting legs extending from an upper end of the barrel body and arranged about a central axis of the barrel body, and the plurality of upper connecting legs and the plurality of retaining arms are arranged alternately about the central axis of the barrel body; and the shaft portion further comprises a plurality of protrusions arranged about the central axis of the shaft portion and protruding outwardly from the shaft body, and the upper stop structure comprises a plurality of upper stop bosses respectively formed by the plurality of protrusions; wherein the plurality of upper connecting legs respectively abut against the plurality of upper stop bosses.

7. The fastener according to claim 6, wherein the receiving barrel further comprises a plurality of lower connecting legs extending from a lower end of the barrel body and arranged about the central axis of the barrel body, and the plurality of lower connecting legs extend inwardly obliquely towards the central axis of the barrel body; and the shaft portion further comprises a plurality of lower recesses arranged about the central axis of the shaft portion and recessed inwardly from the shaft body, and the lower stop structure comprises a plurality of lower stop bosses respectively formed by the plurality of lower recesses; wherein the plurality of lower connecting legs respectively abut against the plurality of lower stop bosses, and the plurality of lower connecting legs are respectively at least partially received by the plurality of lower recesses.

8. The fastener according to claim 7, wherein the receiving barrel further comprises a plurality of guiding tabs extending from the lower end of the barrel body and arranged about the central axis of the barrel body, the plurality of guiding tabs extending inwardly obliquely towards the central axis of the barrel body, and the plurality of guiding tabs and the plurality of lower connecting legs being alternately arranged about the central axis of the barrel body.

9. The fastener according to claim 8, wherein the barrel body has a hexagonal cross section, and the receiving barrel comprises three lower connecting legs, three guiding tabs, three upper connecting legs and three retaining arms, wherein the three lower connecting legs and the three guiding tabs respectively extend downwardly from six sides of the barrel body, and the three upper connecting legs and the three retaining arms respectively extend upwardly from the six sides of the barrel body.

10. The fastener according to claim 8, wherein the receiving barrel has a break enabling the receiving barrel to expand in the radial direction of the receiving barrel.

11. The fastener according to claim 8, wherein the shaft portion comprises a plurality of upper ribs located above the upper stop structure, and the plurality of upper ribs being arranged about the central axis of the shaft portion and cooperating with the hole to center the fastener in the hole; and wherein the shaft portion further comprises a plurality of receiving recesses formed between adjacent ones of the plurality of upper ribs, and the plurality of receiving recesses are configured to respectively receive the plurality of retaining arms when the plurality of retaining arms are deflected towards the shaft portion.

12. The fastener according to claim 11, wherein the plurality of retaining arms have a substantially V shape such that each of the plurality of retaining arms has a lower portion extending obliquely away from the central axis of the barrel body from the barrel body and an upper portion extending obliquely inwardly toward the central axis of the barrel body from the lower portion, and the upper portion has a contact portion for abutting against the hole.

13. The fastener according to claim 12, wherein the end of the upper portion is bent outwardly to form a limiting stop retaining the upper portion in the hole.

14. The fastener according to claim 13, wherein the plurality of retaining arms have arcuate protrusions on outer surfaces thereof.
